

Exploring Key Performance Indicators for Thermal Resilience in Canadian Multi-Unit Residential Buildings

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ABSTRACT

In the past few decades, Canadian climate zones have seen their temperatures increase at rates that are double to triple the global average, resulting in an increase in the frequency, duration, and intensity of extreme heat events. The Canadian building stock – especially multi-unit residential buildings (MURBs) – has demonstrated poor thermal resilience in the face of several such events in the past decade. Given that this upward trend of global temperature is projected to continue under every conceivable future climate scenario, it is critical that the design and construction of new Canadian MURBs considers thermal resilience for the changing climate that they will operate in. Building designers and stakeholders will need key performance indicators (KPIs) to help quantify and characterize the thermal resilience of their buildings. While recent efforts such as IEA EBC Annex 80 Resilient Cooling of Buildings have identified KPIs for thermal resilience, guidance on their use, recommended ranges, and the underlying assumptions used in their formulation (e.g., occupant behaviour, weather files, thermal preferences, etc.) have been largely untested in the context of Canadian MURBs. This paper aims to explore the impact of these phenomena on the different thermal resilience KPIs through a simulation exercise. A Canadian MURB archetype is modelled using the Building Technology Assessment Platform (BTAP). The results will help inform the impact different assumptions have on the resulting KPIs and provide guidance for the use of specific KPIs for evaluating the thermal resilience of Canadian MURBs under different climate, occupancy, or extreme heat event scenarios.

INTRODUCTION

Temperatures around the globe have and will continue to increase over the remainder of the twenty-first century under every conceivable future climate scenario (Gaur and Lacasse, 2022). The threats posed by a warming climate – such as extreme heat and weather events – will challenge the resilience of the built environment. This challenge has and will continue to be uniquely difficult in Canada, where the rates of temperature increase vary from nearly double the global average at its southernmost border to over triple the global average in parts of its Arctic and Subarctic regions (ECCC, 2024). A building performance simulation (BPS) -based approach for evaluating the thermal resilience of the built environment is critical given the threat our changing climate poses. Such approaches are risk-free as occupants are simulated, and are flexible as different weather conditions, buildings, and technologies can be rapidly and dynamically prototyped and evaluated. However, while many standards and guidelines exist around the use of BPS for the purpose of evaluating building energy use, the same standardization around evaluating building thermal resilience is lacking.

The recently completed International Energy Agency (IEA) Energy in Buildings and Communities (EBC) Annex 80 (Holzer, 2024) titled *Resilient Cooling of Buildings* recognized the challenges above and – in an effort to help their participants from 33 institutions spread across 18 countries compare different resilient cooling techniques and technologies – developed a

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simulation guideline (Zhang et al., 2023). This guideline outlines specific weather file formats (Machard et al., 2020), reference building characteristics to be reported, simulations cases, and critically, key performance indicators (KPIs) (Holzer et al., 2024) for evaluating the effectiveness of different resilient cooling strategies in buildings. These include KPIs related to thermal comfort, heat stress, energy performance, and the impact on the operation of heating, ventilation, and air-conditioning (HVAC) equipment and the grid. While a report on a collection of simulations performed in various climates using various resilient cooling technologies should be forthcoming from the Annex participants, currently, a limited set of simulation results are publicly available (Qi et al., 2024) from case study buildings located in ASHRAE Climate Zones 6A (three in Montréal, Canada, and one in Gävle, Sweden), 4A (Leuven, Belgium), and 0B (Rajkot, India) (ASHRAE, 2020). Outside of these case studies, a large series of simulations which explore the performance of archetypical buildings across a wide range of climate zones using the simulation guideline has not been performed to date, including in a Canadian context. Such a study may be of interest to find more generalizable techniques and technologies to enhance thermal resilience that can be applied to the Canadian building stock.

To enable this assessment of the Canadian building stock, archetype reference models are needed. Currently, Canadianized equivalents (Kirney et al., 2023) of the U.S. Department of Energy (DOE) commercial reference buildings (Deru et al., 2011) which utilize National Energy Code of Canada for Buildings (NECB) (CCBFC, 2022) requirements exist as part of the Building Technology Assessment Platform (BTAP) (NRCan, 2025a). Therefore, the aim of this study is to pair BTAP and its Canadian commercial reference buildings with the workflow outlined in the Annex 80 simulation guideline to determine baseline KPI values for these reference buildings, starting with a midrise multi-unit residential building (MURB). This will lay the groundwork for future work to explore reference building KPIs for the other archetypes, and the eventual expansion of BTAP's open-source code to allow for thermal resilience KPIs and resilient cooling techniques and technologies to be integrated into its functionality natively. The data from this study will also be structured in the format outlined in the simulation guideline so that it may be contributed to the growing body of data and literature utilizing this simulation approach.

The remainder of this paper is structured as follows. First, the methodology, including BTAP, the overall simulation approach, the reference buildings, and the KPIs considered are introduced. Then, the results of the simulations of the midrise MURB are presented, and the impact of different simulation assumptions on the KPIs as well as the studies limitations are discussed. Then, the conclusions are presented, followed by the identification of areas where future work is needed.

METHODOLOGY

Building Technology Assessment Platform (BTAP)

openstudio-standards (NREL, 2024) is an open-source library which uses the OpenStudio® SDK (NREL, 2025) – a software development kit (SDK) and command line interface (CLI) for programmatically accessing EnergyPlus™ modelling engine – to automatically generate OpenStudio® models (OSMs) given building geometry and a reference building code. BTAP is the Canadian contribution to *openstudio-standards* which allows for these models to be generated based on the requirements of the NECB (Kirney et al., 2023).

In order to orchestrate the programmatic creation and running of OSMs created through the BTAP portion of *openstudio-standards*, the open-source CLI-based tool *btap_batch* (NRCan, 2025a) can be used. This run manager allows large sets of simulations to be performed locally or through cloud computing; available analyses include optimization using a genetic algorithm, parametric analysis, sensitivity analysis, and Latin hypercube sampling (LHS) for large solution spaces. The user provides an input file including (but not limited to) the analysis type, building archetypes, versions of NECB, weather files (NRCan, 2025b), systems (e.g., service hot water (SHW) or HVAC) and their fuels, and various energy conservation measures (ECMs) such as envelope properties and load manipulations of interest. *btap_batch* then automatically creates, runs, and outputs the results (including the individual models as well as a summary) of the simulations. This workflow allows for the rapid creation and analysis of archetypical models representing the Canadian commercial building stock developed by Kirney et al. (2023). One such model – the midrise apartment archetype – was utilized in this study and is described in the following sections.

Simulation Approach

The simulation guideline (Zhang et al., 2023) calls for a specific set of simulations to be performed depending on whether

or not mechanical cooling is considered to be available in the building of interest. As the BTAP midrise apartment archetype has mechanical cooling per NECB, there are five baseline (i.e., reference) cases which must be simulated using four separate weather files: three typical meteorological year (TMY) files and one mandatory heatwave year (HWY) file. The specific weather files are developed using the methodology described in Marchard et al. (2020, 2024) to predict future weather out to the year 2100 based on regional climate model simulations using the worst-case scenario representative concentration pathway (RCP8.5). The three TMYs represent (1) the contemporary period (i.e., approximately 2010), (2) the medium-term future (i.e., approximately 2050), and (3) the long-term future (i.e., approximately 2090); these files are each simulated once for an entire year (i.e., 8760 hours) under normal operation. The mandatory HWY file is created by selecting the year with the most severe heatwave (i.e., the largest average temperature increase for the duration of the heatwave); this file is simulated twice for a duration spanning one week before and one month after the heatwave: once under normal operation, and once with a simulated power outage (i.e., no mechanical cooling available). Note that the timing and length of the heatwave may vary from location to location.

The locations selected were the three largest Canadian cities based on the availability of these weather files (Zou et al., 2022) which are available on the Annex 80 website (IEA, 2025) in the EnergyPlus™ weather (EPW) format: Vancouver, British Columbia (BC) (ASHRAE Climate Zone 4C), Toronto, Ontario (ON) (ASHRAE Climate Zone 5A), and Montréal, Québec (QC) (ASHRAE Climate Zone 6A). Note that BTAP requires design day information (DDY) and EnergyPlus™ weather statistic (STAT) files; these files were generated for each EPW using the EnergyPlus™ weather converter pre-processing utility (v8.1.0) included with EnergyPlus™ (v24.2.0) (DOE, 2024a). The heatwave periods for the HWY files are summarized in Table 1. Note that the *PowerOutage* measure developed by Goldwasser et al. (2022) was integrated into a custom branch of *openstudio-standards* used for this study, which was called by *btap_batch* to simulate outages during the heatwave periods.

Table 1. Heatwave Durations (Zou et al., 2022)

Location	Heatwave start (yyyy-mm-dd)	Heatwave end (yyyy-mm-dd)	Duration (days)
4C (Vancouver, BC)	2087-07-28	2087-08-23	27
5A (Toronto, ON)	2091-06-29	2091-08-24	57
6A (Montréal, QC)	2090-07-07	2090-07-28	22

The midrise apartment archetype based on the 2020 version of NECB (CCBFC, 2022) was simulated in these three locations for a total of 15 simulations (i.e., one archetype, one version of NECB, three TMYs under normal operation, one HWY under normal operation and during a power outage, and three locations) using the parametric analysis functionality of *btap_batch*.

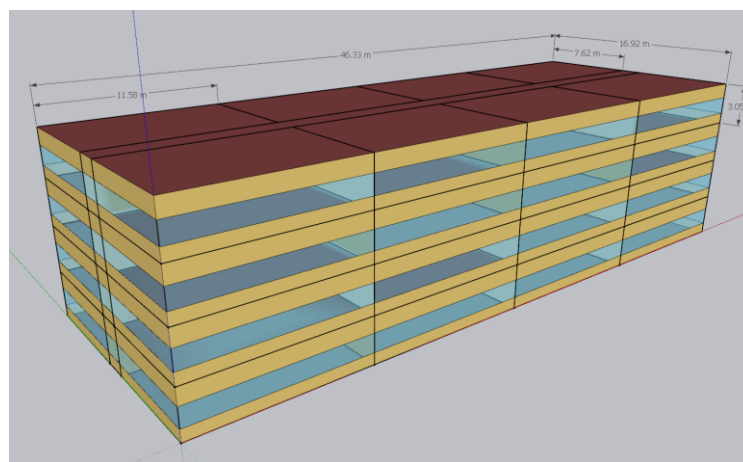


Figure 1 Example of the BTAP midrise apartment archetype geometry and massing. Note that the building is oriented along cardinal directions; the longitudinal axis of the building runs in the east-west direction.

Midrise Apartment Archetype

The midrise apartment archetype, depicted in Figure 1, is based on the building geometry created by the U.S. Department of Energy (DOE) for their commercial reference building models (Deru et al., 2011). The geometry and characteristics of this archetype are further summarized in Table 2. Note the slight variations in the vertical fenestration and door area to gross wall area ratio (FDWR) as well as the heating system fuel types between climate zones; these differences reflect the requirements laid out in NECB 2020 (CCBFC, 2022). Briefly, the building consists of four stories. Each story consists of a central double-loaded corridor with four identically sized dwelling units on each side, for a total of eight occupied units per floor and 32 dwelling units in the building. Each dwelling unit and corridor is treated as its own thermal zone, for a total of 36 thermal zones (i.e., 32 dwelling units and four corridors). Ventilation and conditioned air are provided to the dwelling units via a centralized dedicated outdoor air system (DOAS) (i.e., system 1 per NECB 2020), while the corridors utilize a centralized make-up air unit (MAU) (i.e., system 4 per NECB 2020), with minimum outdoor ventilation air rates commensurate with ASHRAE 62.1-2016 (ASHRAE, 2016) requirements.

Table 2. Building Geometry and Characteristics

Characteristic	Midrise Apartment		
	4C (Vancouver, BC)	5A (Toronto, ON)	6A (Montréal, QC)
Floors	4	4	4
Conditioned floor area	3135 m ² (33,740 ft ²)	3135 m ² (33,740 ft ²)	3135 m ² (33,740 ft ²)
Footprint	784 m ² (8435 ft ²)	784 m ² (8435 ft ²)	784 m ² (8435 ft ²)
Floor-to-ceiling height	3 m (10 ft)	3 m (10 ft)	3 m (10 ft)
FDWR	40%	40%	38.7%
Thermal zones	One double-loaded corridor zone with eight perimeter dwelling unit zones per floor	One double-loaded corridor zone with eight perimeter dwelling unit zones per floor	One double-loaded corridor zone with eight perimeter dwelling unit zones per floor
Heating system	Natural gas-fired boiler with hydronic baseboards	Natural gas-fired boiler with hydronic baseboards	Electric boiler with electric resistance baseboards
Cooling system	Central air-cooled DX	Central air-cooled DX	Central air-cooled DX

The midrise apartment archetype utilizes hourly schedules for its plug-in equipment, lighting, HVAC operation (i.e., fan on/off, heating/cooling setpoints), SHW usage, and occupancy based on the requirements of NECB 2020, namely Table A-8.4.3.2.(1)-J (CCBFC, 2022). Note that the corridors utilize the same schedule based on NECB recommendations. The recommended load values for each space type per NECB, summarized in Table 3, are multiplied by the fractional schedule to determine the load in each zone at each hour. The three TMY simulations and single HWY simulation under normal operation are simulated with these loads. For the single HWY simulation during a power outage, the plug-in equipment, lighting, HVAC operation, and SHW schedules are reduced to zero for the duration of the outage for residential buildings per the simulation guideline (Zhang et al., 2023). However, the occupancy schedule remains unchanged for all simulations.

Table 3. Building Loads

Load	Space Type	
	Dwelling Unit	Corridor
Occupant density (m ² /occupant)	25	100
Plug-in equipment load (W/m ²)	5	0
Service hot water load (W/occupant)	500	0
Lighting levels (lx)	300	150

Key Performance Indicators (KPIs)

The KPIs used in the simulation guideline are selected from the list of KPIs developed as part of the Annex (Holzer et al., 2024) and are described in full in the guideline itself (Zhang et al., 2023). Briefly, the first KPIs for thermal resilience considered is indoor overheating degree (IOD), see Equation 1 (Hamdy et al., 2017):

$$IOD = \frac{\sum_{z=1}^Z \sum_{i=1}^{N_{occ}(z)} [(T_{op,i,z} - TL_{comf,i,z})^+ \cdot t_{i,z}]}{\sum_{z=1}^Z \sum_{i=1}^{N_{occ}(z)} t_{i,z}} \quad (1)$$

where T_{op} is the operative temperature (°C), TL_{comf} is the operative temperature limit (°C), N_{occ} is the number of occupied hours, z and i are the zone and occupied-hour indices, and t is the time step (i.e., 1 h). Note that a fixed TL_{comf} of 26 °C considered in this study per CIBSE Guide A (CIBSE, 2015); this is the lower limit for a bedroom area as the dwelling units in the midrise archetype are modelled as a single zone and assumes no ceiling fan is present. The threshold of 26 °C is also currently being considered for dwelling units in the forthcoming version of the NECB (CBHCC, 2025).

The second KPI for thermal resilience, ambient warmth degree (AWD), is described by Equation 2 (Hamdy et al., 2017):

$$AWD_b = \frac{\sum_{i=1}^{N_{occ}} [(T_{a,i} - T_b)^+ \cdot t_i]}{\sum_{i=1}^{N_{occ}} t_i} \quad (2)$$

where T_a is the outdoor air dry-bulb temperature (°C) and T_b is the base dry-bulb temperature (°C). Note that the base temperature is set at 18 °C per the simulation guideline.

The third KPI for thermal resilience, climate change overheating resistivity (CCOR), combines the previous two KPIs, see Equation 3 (Rahif et al., 2022):

$$\frac{1}{CCOR} = \frac{\sum_{Sc=1}^{Sc-M} (IOD_{Sc} - \overline{IOD}) \times (AWD_{Sc} - \overline{AWD})}{\sum_{Sc=1}^{Sc-M} (AWD_{Sc} - \overline{AWD})^2} \quad (3)$$

where Sc is the climate scenario index and M is the total number of climate scenarios considered. Note that CCOR is calculated per building and climate zone; one CCOR value was calculated per climate zone for the midrise apartment archetype using the three TMY climate scenarios (i.e., $M = 3$).

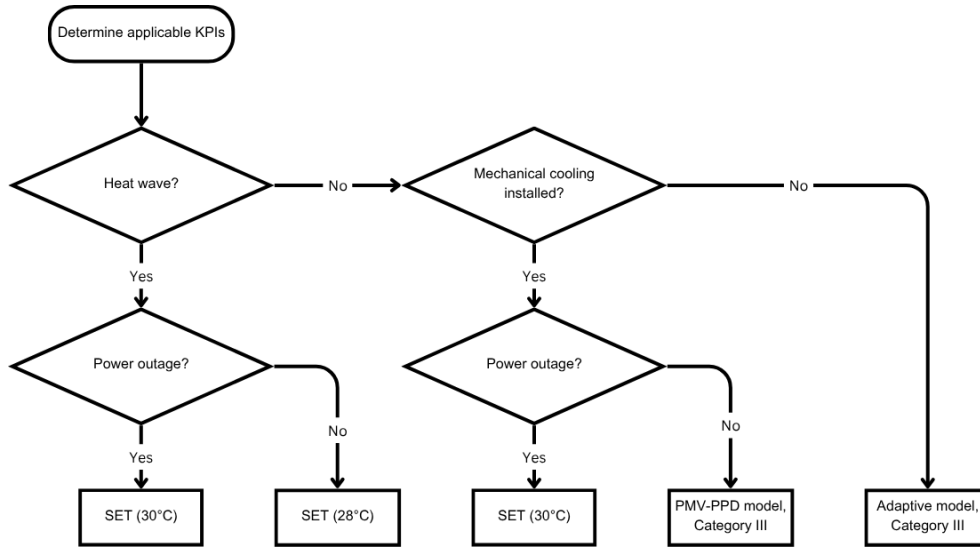


Figure 2 Thermal comfort and heat stress model selection guidance based on the simulation guideline (Zhang et al., 2023).

With respect to thermal comfort and heat stress, the simulation guideline provides guidance on selecting KPIs depending on the scenario considered, see Figure 2. However, EnergyPlus™ uses the ASHRAE Standard 55-2004 (ASHRAE, 2004) simple model as the thermal comfort model by default (DOE, 2024b), which does not provide the required outputs to compute each KPI. As such, a new branch of *openstudio-standards* was created to enable the *Fanger*, *Pierce*, and *AdaptiveCEN15251* comfort models and their outputs (i.e., PMV, PPD, SET, and adaptive model status); these models are described in the EnergyPlus™ documentation (DOE, 2024b). This branch of *openstudio-standards* was called upon by *btap_batch* for each simulation. Note that the BTAP midrise archetype model contains default assumptions for inputs required for these models based on the NECB, namely a clothing insulation level of 0.5 clo from May 1st to September 30th inclusive, and 1.0 clo for the remainder of the year. The occupant activity schedule uses a constant level of 130 W (i.e., just above resting levels) for occupants in dwelling units. The air velocity experienced by the occupants is assumed to be constant at 0.2 m/s (i.e., the upper air speed expected in a well-mixed zone without additional equipment such as fans (ASHRAE, 2023)).

In order to quantify thermal comfort, the predicted mean vote- (PMV-) predicted percentage dissatisfied (PPD) model was used as specified by the guideline for a mechanically cooled building under normal operating conditions (Zhang et al., 2023). This calculation method multiplies the number of hours where the PMV exceeds 0.7 (i.e., Category III of ISO 17772-1 (ISO, 2017)) by the PPD to adjust for the level of dissatisfaction experienced by occupants. This thermal comfort KPI is described in Equation 4:

$$PPD_{PMV} = \sum_{z=1}^Z \sum_{i=1}^{N_{occ}(z)} PPD_{i,z} n_{i,z} \delta_{i,z} \quad \text{where} \quad \delta_{i,z} = \begin{cases} 1 & |PMV_{i,z}| > PMV_{lim} \\ 0 & |PMV_{i,z}| \leq PMV_{lim} \end{cases} \quad (4)$$

where PPD_{PMV} is the PMV-weighted PPD, n is the number of occupants, and PMV_{lim} is 0.7 per Category III of ISO 17772-1 (ISO, 2017). Notably, this is the only thermal comfort metric from Holzer et al. (2024) required by the simulation guideline for mechanically cooled buildings; other metrics are considered optional, while other scenarios require KPIs related to heat stress.

In order to quantify heat stress, the number of zone hours where the standard effective temperature (SET) exceeded 28 °C ($SET_{28^\circ C}$) and 30 °C ($SET_{30^\circ C}$) were calculated for the period of the heatwave (recall Table 1) for the operating case and outage case, respectively, per the simulation guideline (Zhang et al., 2023). This relationship is described in Equation 5:

$$SET_{ref} = \sum_{z=1}^Z \sum_{i=1}^{N_{occ}(z)} \delta_{i,z} \quad \text{where} \quad \delta_{i,z} = \begin{cases} 1 & SET_{i,z} > SET_{lim} \\ 0 & SET_{i,z} \leq SET_{lim} \end{cases} \quad (5)$$

where SET_{ref} is either $SET_{28^\circ C}$ or $SET_{30^\circ C}$ and SET_{lim} is either 28 °C or 30 °C SET, respectively. Note that both the thermal comfort and heat stress metrics are only calculated for occupied hours; the building is considered to be occupied 24/7 (recall the NECB 2020 occupancy schedule in NECB 2020 Table A-8.4.3.2.(1)-J (CCBFC, 2022)).

The remaining KPIs for the simulation guideline relate to energy use and emissions. The primary energy use of the HVAC system per annum, normalized by the conditioned floor area (kWh/m²·a) is reported from each simulation, as well as the associated CO₂ equivalent (kgCO₂e/m²·a) and the emission factors (EFs) used in their calculation. The simulation guideline does not provide any recommendations for EF forecasts over the duration of the TMY periods considered; as these climate scenarios are based on the worst-case business-as-usual emissions scenarios, contemporary EFs for the three locations are used for each simulation. BTAP contains built-in annual average EFs for each fuel type (i.e., natural gas and electricity) and province/territory based on data from the U.S. Environmental Protection Agency (EPA) (EPA, 2020). The values used for the locations studied are summarized in Table 4. The KPIs were determined via post-processing of the hourly outputs from each simulation using Python. Work is underway to integrate the calculation of these KPIs directly in *btap_batch* such that no post-processing is required.

Table 4. BTAP Annual Emission Factors (EPA, 2020)

Location	Emission Factor (gCO ₂ e/kWh)	
	Natural Gas	Electricity
Vancouver, BC	181.5	15.3
Toronto, ON	177.9	36.0
Montréal, QC	177.8	1.8

RESULTS & DISCUSSION

The KPIs computed for the midrise apartment archetype under the simulation scenarios considered are summarized in Table 5. The CCOR values indicate that mechanically cooled midrise apartments built to NECB 2020 requirements should be able to suppress the increasing outdoor thermal stress due to climate change when operating for the climate zones studied (i.e., all CCOR > 1) when examined using the simulation guideline (Zhang et al., 2023), with Toronto, ON (ASHRAE Climate Zone 5A) having the highest resistance compared to Montréal, QC (ASHRAE Climate Zone 6A) and the lowest resistance in Vancouver, BC (ASHRAE Climate Zone 4C). However, given magnitude of the CCORs over the resistivity threshold, the order of locations studied is negligible. When considering mechanical cooling, the IOD values remain considerably low; this is due to the comfort limit (TL_{comf}) of 26 °C being higher than the zone air temperature setpoint in NECB 2020 (i.e., 24 °C), resulting in operative temperatures (T_{op}) in the zones that are frequently below the comfort limit. For this reason, the IOD – and by extension the CCOR – provide limited insights into the performance of these mechanically cooled buildings. While the baseline values in Table 5 may be a point of comparison for when assessing other active cooling technologies, the utility of this comparison is limited given the magnitude by which a mechanically cooled building exceeds the CCOR threshold. While not examined directly in this study, the use of these metrics for naturally ventilated buildings would inherently have higher utility given the expected rise in outdoor air temperatures and how closely a naturally ventilated building's zone air temperatures may track with these outdoor temperatures without any passive measures.

Table 5. KPI Values

Location	EPW	IOD (°C)	AWD _{18°C} (°C)	CCOR	Model	Duration (days)	Value	Energy Use (kWh/m ² ·a)	Emissions (kgCO ₂ e/m ² ·a)
Vancouver, BC	2010 TMY	0.09	1.02	115.3	PPD _{PMV}	165	3.8	87.4	8.62
	2050 TMY	0.09	1.50		PPD _{PMV}	179	4.2	82.8	7.48
	2090 TMY	0.10	2.02		PPD _{PMV}	199	4.5	81.6	6.77
	Operating HWY	0.12	2.41	-	SET _{28°C}	27	4.7	79.9	6.08
	Outage HMY		2.41	-	SET _{30°C}	27	522	74.0	5.97
Toronto, ON	2010 TMY	0.06	1.02	171.1	PPD _{PMV}	178	3.47	77.3	7.56
	2050 TMY	0.06	1.58		PPD _{PMV}	202	3.53	79.7	7.58
	2090 TMY	0.07	2.26		PPD _{PMV}	205	3.9	75.7	6.41
	Operating HWY	0.07	2.17	-	SET _{28°C}	55	7.8	75.4	6.50
	Outage HMY		2.17	-	SET _{30°C}	55	370	64.1	6.41
Montréal, QC	2010 TMY	0.05	1.09	142.3	PPD _{PMV}	177	3.55	74.4	0.13
	2050 TMY	0.06	1.50		PPD _{PMV}	185	3.85	73.8	0.13
	2090 TMY	0.06	1.82		PPD _{PMV}	199	3.95	68.8	0.12
	Operating HWY	0.08	2.33	-	SET _{28°C}	22	7.4	73.4	0.13
	Outage HMY		2.33	-	SET _{30°C}	22	432	69.0	0.13

The thermal comfort and heat stress KPIs calculated by the different models highlight a pattern in the different weather files (Zou et al., 2022) used for the 2010, 2050, and 2090 periods. In each case, the AWD indicates that the outdoor temperatures for each subsequent representative TMY are increasing. At the same time, the PPD_{PMV} is calculated for the cooling season in the building, which varies between locations and TMY file. For the purpose of this study, the first and last hour that the *AdaptiveCEN15251* model status increases above -1 (i.e., when outdoor air is considered warm enough to be viable for natural ventilation) is considered the beginning and the end of the cooling season; no alternative guidance on the consideration of the cooling season duration is provided in the simulation guideline. While the PPD_{PMV} has been normalized by this duration, the usage of a single metric to quantify the PPD_{PMV} across all zones in the building should further necessitate the normalization of metrics by occupancy; for example, a highrise apartment archetype would inherently have higher PPD_{PMV} than the midrise apartment archetype as calculated. While individual zone-level metrics could be reported for each zone, this volume of information would make deriving insights from the reported metrics less accessible, especially in a building code context.

Because NECB 2020 states a single schedule for each occupancy type, reporting a range of individual values for each comfort model or heat stress metric experienced by the zone types simulated may present a better alternative to the bulk metrics currently employed. Further guidance on thresholds to determine the start and stop of the cooling season and occupant normalized metrics may be warranted to facilitate equitable comparisons between buildings, climate zones, and weather files.

Similarly to the CCOR and IOD, the number of SET_{28°C} exceedances per day was trivially low in the mechanically cooled building for the operating HWY scenario. However, the SET_{30°C} values were significant during the outage HWY. The 36-zone building had a maximum possible number of exceedance hours of 864 h per day; during the outage periods, the midrise apartment archetype averaged between 10.3 h to 14.5 h per day per zone. Notably, these values assume that the building does not undergo any natural ventilation during this period (i.e., windows are not operated despite the mechanical cooling equipment failing). Considerations for natural ventilation would likely decrease the number of hours spent at these elevated temperatures, although the simulation guideline does not provide guidance on mixed-mode buildings or alternative modes of operations during emergencies. Similarly, the guideline does not provide recommendations on window usage behaviour. Considerations should be made for how occupants would realistically use windows during such a heat event; this has been the subject of extensive study to date. Beyond occupant window usage behaviours, other occupant behaviours (such as occupancy patterns, clothing, and activity levels) should be further examined during heatwave periods to address the impact these changes may have on these metrics.

CONCLUSION

In this study, new branches of *openstudio-standards* and *btap_batch* were created to enable simulation outputs and scenarios needed to compute thermal resilience KPIs from the simulation guideline (Zhang et al., 2023) produced by *IEA EBC Annex 80 Resilient Cooling of Buildings* (Holzer, 2024). The utility of the simulation approach and KPIs themselves were evaluated on a mechanically cooled midrise apartment archetype model in Vancouver, BC (ASHRAE Climate Zone 4C), Toronto, ON (ASHRAE Climate Zone 5A), and Montréal, QC (ASHRAE Climate Zone 6A). The results included baseline KPI values for this archetype in these climate zones and identified several strengths and weaknesses of these KPIs and the simulation approach in the Canadian context, namely the utility of the IOD, CCOR, and SET_{28°C} for mechanically cooled buildings, clarifications around the periods used for time-normalized metrics and normalization of metrics by occupancy to allow for equitable comparison, as well as assumptions about occupant behaviour. The findings from this study will be used to inform next steps as Canadian codes seek to meaningfully quantify the thermal resilience of our built environment. Future work will leverage the branches developed for this study to directly calculate these KPIs as part of *btap_batch* outputs natively, allowing for other locations, archetypes, and KPIs to be examined rapidly and programmatically.

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NOMENCLATURE

M	=	number of climate scenarios	$comf$	=	comfort
N	=	number	i	=	occupied hour index
n	=	number of occupants	lim	=	limit
T	=	temperature	occ	=	occupied hours
TL	=	operative temperature limit	op	=	operative
Z	=	total number of zones	ref	=	reference
			Sc	=	climate scenario index
			t	=	timestep index
			z	=	zone index

Subscripts

a	=	outdoor air dry-bulb
b	=	base dry-bulb temperature

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